

SALES PERFORMANCE ANALYSIS REPORT

Project Title

Sales Performance Analysis Using Superstore Sales Dataset

Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Power BI (Dashboard)

Dataset Reference

<https://www.kaggle.com/datasets/bhanupratapbiswas/superstore-sales>

1. Executive Summary

This project focuses on analyzing retail sales data to understand business performance across different regions, product categories, customer segments, and time periods. The analysis aims to identify key sales trends, evaluate product performance, detect seasonal demand patterns, and generate actionable business recommendations.

The dataset was cleaned, transformed, and analyzed using Python. Various visualizations including bar charts, pie charts, histograms, line charts, box plots, and heatmaps were created to gain insights into sales behavior. The cleaned dataset was also prepared for dashboard development in Power BI.

2. Project Objectives

The primary objectives of this project are:

- Analyze sales performance across regions.
 - Analyze sales performance across product categories and sub-categories.
 - Identify best-performing and worst-performing products.
 - Examine customer purchasing behavior.
 - Study shipping preferences and delivery performance.
 - Identify monthly, quarterly, and yearly sales trends.
 - Detect seasonality in sales.
 - Provide tactical business recommendations for improving sales performance.
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3. Dataset Description

The dataset contains historical sales transactions from a retail superstore.

Attributes Used in Analysis

- Row_ID
- Order_ID
- Order_Date
- Ship_Date
- Ship_Mode
- Customer_ID
- Customer_Name
- Segment
- Country
- City
- State
- Postal_Code
- Region
- Product_ID
- Category
- Sub_Category
- Product_Name
- Sales

These attributes provide information related to customers, products, orders, locations, shipping methods, and sales revenue.

4. Data Cleaning and Preparation

Data preprocessing was performed to ensure data quality and consistency before analysis.

4.1 Date Conversion

Order_Date and Ship_Date columns were converted into datetime format to enable time-based analysis.

4.2 Duplicate Removal

Duplicate records were identified and removed from the dataset.

4.3 Missing Value Inspection

The dataset was checked for null values to ensure completeness and reliability.

4.4 Feature Engineering

Additional columns were created to support advanced analysis:

Year

Used for yearly sales trend analysis.

Month

Used for monthly sales analysis.

Month_Name

Used to identify peak sales months.

Quarter

Used for quarterly trend analysis.

Day_Name

Used to identify sales patterns by day.

Shipping_Days

Calculated using:

Shipping Days = Ship Date – Order Date

This metric helps evaluate delivery performance.

5. Cleaned Dataset Creation

After cleaning and feature engineering, a new dataset named:

Superstore_Cleaned.csv

was generated and saved for future analysis and Power BI dashboard development.

This ensures consistency between Python-based analysis and dashboard visualizations.

6. Key Performance Indicators (KPIs)

Several KPIs were calculated to evaluate business performance.

Total Revenue

Represents the total sales generated across all orders.

Formula:

Total Revenue = Sum of Sales

Total Orders

Represents the total number of unique orders.

Formula:

Total Orders = Count of Unique Order_ID

Unique Customers

Represents the number of distinct customers.

Formula:

Unique Customers = Count of Unique Customer_ID

Average Order Value (AOV)

Measures average revenue generated per order.

Formula:

Average Order Value = Total Revenue / Total Orders

Average Customer Value

Measures average revenue generated per customer.

Formula:

Average Customer Value = Total Revenue / Unique Customers

Limitation

Profit Margin and Conversion Metrics were not calculated because the dataset does not contain Profit, Cost, Lead, Visitor, or Conversion information.

7. Sales Distribution Analysis

A histogram with KDE was used to analyze sales distribution.

Purpose

- Understand sales transaction patterns.
- Identify high-value orders.
- Detect outliers.

Findings

- Most transactions occur within lower sales ranges.
 - A smaller number of transactions generate exceptionally high sales.
 - Sales distribution is positively skewed.
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8. Regional Sales Performance Analysis

Sales performance was analyzed across different regions.

Visualizations Used

- Bar Chart
- Pie Chart

Purpose

- Compare revenue contribution among regions.
- Identify strongest and weakest markets.

Insights

- Regional sales vary significantly.
 - Some regions contribute a larger share of revenue.
 - High-performing regions demonstrate stronger customer demand.
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9. State-Level Performance Analysis

Top-performing states were identified using total sales.

Visualization Used

Horizontal Bar Chart

Purpose

- Identify top revenue-generating states.
- Support regional expansion planning.

Insights

- A limited number of states contribute a significant share of revenue.
 - Underperforming states represent future growth opportunities.
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10. Category Performance Analysis

Products were grouped into major categories.

Visualizations Used

- Bar Chart
- Pie Chart

Purpose

- Evaluate category contribution to revenue.
- Understand customer preferences.

Insights

- Revenue is concentrated within specific categories.
 - High-performing categories should receive greater business focus.
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11. Sub-Category Analysis

Sales performance was analyzed at a more detailed level.

Visualization Used

Bar Chart

Purpose

- Identify top-performing product groups.

- Improve inventory planning.

Insights

- Certain sub-categories dominate sales.
 - High-demand sub-categories present opportunities for growth.
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12. Product Performance Analysis

Individual products were evaluated based on total sales.

Best Performing Products

The top 10 products were identified based on sales revenue.

Business Value

- High customer demand.
- Significant contribution to total revenue.
- Strong candidates for promotional campaigns.

Worst Performing Products

The bottom 10 products were identified.

Business Value

- Require strategic review.
 - May need discounts or repositioning.
 - Potential candidates for discontinuation.
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13. Monthly Sales Trend Analysis

Monthly sales trends were analyzed using a line chart.

Purpose

- Detect seasonality.
- Identify peak sales periods.

Insights

- Sales fluctuate across months.
 - Certain months consistently outperform others.
 - Seasonal demand patterns are evident.
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14. Quarterly Sales Trend Analysis

Quarterly performance was analyzed to identify broader trends.

Purpose

- Evaluate business performance by quarter.
- Support strategic planning.

Insights

- Revenue varies across quarters.
 - High-performing quarters contribute significantly to annual sales.
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15. Yearly Sales Trend Analysis

Year-over-year performance was analyzed.

Purpose

- Measure long-term business growth.

Insights

- Revenue trends indicate overall business performance over time.
 - Useful for forecasting and strategic decision-making.
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16. Customer Segment Analysis

Sales were analyzed by customer segment.

Visualization Used

Bar Chart

Purpose

- Understand purchasing behavior.
- Identify the most valuable customer groups.

Insights

- Some customer segments contribute more revenue than others.
 - Segment-specific strategies can improve customer retention.
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17. Shipping Mode Analysis

Sales performance was evaluated across shipping methods.

Visualization Used

Pie Chart

Purpose

- Identify preferred shipping options.
- Understand customer delivery preferences.

Insights

- Certain shipping modes are more frequently used.
 - Shipping preferences influence customer satisfaction.
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18. Shipping Days Analysis

Delivery time was analyzed using a box plot.

Purpose

- Evaluate shipping efficiency.
- Detect unusual delivery delays.

Insights

- Most deliveries occur within an expected timeframe.
 - Outliers indicate orders with extended delivery durations.
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19. Top Customers Analysis

The top 10 customers were identified based on sales contribution.

Purpose

- Identify high-value customers.
- Support customer retention initiatives.

Insights

- A small number of customers contribute significantly to total revenue.
 - Retaining these customers is important for sustained growth.
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20. Heatmap Analysis

A heatmap was created using Category and Region.

Purpose

- Compare category performance across regions.
- Identify high-performing combinations.

Insights

- Sales performance varies across category-region combinations.
 - Opportunities exist for region-specific product strategies.
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21. Automated Business Insights

The notebook automatically identifies:

- Best Region

- Best Category
- Best Product
- Best Customer Segment
- Peak Sales Month

These insights provide a quick summary of the strongest business drivers.

22. Tactical Recommendations for Alfido Tech

Recommendation 1: Focus on High-Selling Products

Increase inventory and marketing investment for products generating the highest sales revenue.

Expected Benefit

- Increased revenue.
 - Improved product availability.
 - Higher customer satisfaction.
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Recommendation 2: Improve Performance of Low-Selling Products

Review pricing, visibility, and promotional strategies for underperforming products.

Expected Benefit

- Better inventory utilization.
 - Reduced dead stock.
 - Improved product profitability.
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Recommendation 3: Strengthen Regional Sales Strategies

Replicate successful sales strategies from top-performing regions in weaker markets.

Expected Benefit

- Increased regional market penetration.
 - Improved overall revenue distribution.
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Recommendation 4: Leverage Seasonal Demand

Prepare inventory and marketing campaigns before peak sales periods.

Expected Benefit

- Increased seasonal revenue.
- Improved demand forecasting.
- Better inventory planning.

Recommendation 5: Enhance Customer Retention

Develop loyalty programs and personalized offers for high-value customers and customer segments.

Expected Benefit

- Increased repeat purchases.
- Improved customer lifetime value.
- Stronger customer relationships.

23. Conclusion

This project successfully analyzed sales performance using transactional retail sales data. Through data cleaning, feature engineering, KPI generation, and multiple visualizations, meaningful insights were obtained regarding regional performance, customer behavior, product demand, shipping preferences, and seasonality.

The findings indicate opportunities for improving inventory planning, regional marketing, customer retention, and product portfolio management. The recommendations provided can help Alfido Tech optimize sales performance and support future business growth.

The cleaned dataset generated during this project can also be used to create an interactive Power BI dashboard for real-time business monitoring and decision-making.